Buzzworthy

Insects and their arthropod cousins are everywhere. You will learn to tell them apart by counting body parts, legs, and antennae, and learn how they help and harm plants, animals, and people.

My name: _____

My partner: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 2	Is It an Insect? Meet the Arthropods
Lesson 2	Fri Nov 6	Insect Body Parts, Life Cycles, and 6 Big Orders
Lesson 3	Fri Dec 11	Hoppers, Bugs, Roaches, and Hiders
Lesson 4	Fri Jan 29	Water Insects, Ancient Insects, and the Key
Lesson 5	Fri Feb 26	Practice Tournament
Lab I1	Before Lesson 3	Who Needs Whom? Insect Relationships (home or an open Friday)
Lab I2	Before Lesson 4	Insects, Health, and Money (home or an open Friday)

At the tournament

A 50-minute station event: partners rotate through timed stations with photos, preserved or live specimens, name each specimen's class and/or insect order, then answer a second question about it, using up to 2 hand lenses and 2 references (field guides and/or 1-inch binders).

Keep every handout for this event behind this divider, in order. Bring this binder to every Field A Fridays meeting.

Buzzworthy: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can tell an insect from other arthropods by its 3 body parts, 6 legs, and 1 pair of antennae.	L1 L2	☐	☐
I can name the 5 arthropod classes (Arachnida, Chilopoda, Crustacea, Diplopoda, Insecta) and give an example of each.	L1 L4 L5	☐	☐
I can label an insect's head, thorax, abdomen, antennae, compound eyes, mouthparts, legs, wings, and spiracles.	L2	☐	☐
I can explain complete metamorphosis (egg, larva, pupa, adult) and incomplete metamorphosis (egg, nymph, adult), and say which one an order uses.	L2 L3 L4	☐	☐
I can identify Coleoptera, Lepidoptera, Hymenoptera, Diptera, Neuroptera, and Siphonaptera from a photo or specimen.	L2 L5	☐	☐
I can identify Orthoptera, Hemiptera, Blattodea, Dermaptera, Phasmatodea, and Psocodea from a photo or specimen.	L3 L5	☐	☐
I can identify Odonata, Ephemeroptera, Trichoptera, Archaeognatha, and Zygentoma from a photo or specimen.	L4 L5	☐	☐
I can match mouthparts (chewing, piercing-sucking, siphoning, sponging) to an insect and what it eats.	L2 L3	☐	☐
I can use a dichotomous key to find an arthropod's class and insect order.	L4 L5	☐	☐
I can describe insect habitats (including ponds and streams) and adaptations like camouflage, mimicry, and warning colors.	L3 L4	☐	☐
I can explain pollination, decomposition, food webs, competition, and symbiosis (mutualism, commensalism, parasitism) with insect examples.	L5 I1	☐	☐
I can name arthropods that spread disease (mosquitoes, fleas, lice, ticks) and ways people stay safe.	L5 I2	☐	☐
I can explain how insects help people (food, honey, silk, dyes, medicine) and harm people (crop pests, house pests), and what integrated pest management means.	I2	☐	☐
I can explain how climate change affects insects and ticks (warmer winters, moving ranges, bad timing with flowers).	I1	☐	☐
I can find an answer in our 2 reference binders quickly and finish each station before time is called.	L1 L5 I2	☐	☐

Is It an Insect? Meet the Arthropods

TODAY I CAN

- I can tell an insect from other arthropods by its 3 body parts, 6 legs, and 1 pair of antennae.
- I can name the 5 arthropod classes (Arachnida, Chilopoda, Crustacea, Diplopoda, Insecta) and give an example of each.
- I can find an answer in our 2 reference binders quickly and finish each station before time is called.

An arthropod (AR-thro-pod) is an animal with jointed legs and a hard outside skeleton called an exoskeleton. Insects are one kind of arthropod. This chart is page 1 of your Ref 1 ID Binder.

Today's steps: run it yourselves (22 minutes)

You need: 16 arthropod photo cards (1 set per pair), 4 Station Sprint cards (photo + question); specimens optional, Lesson 1 answer card. Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Take turns reading the intro and 'The 5 arthropod classes' chart out loud. Say the rule: insects have 3 body parts, 6 legs, and 1 pair of antennae.
2. Both 6 min. Set the timer for 6 minutes. Sort your 16 photo cards into 5 piles using the class chart. Count legs, body parts, and antennae out loud before you place each card.
3. Event Captain 2 min. Set the timer for 2 minutes. Read the card sort answers from the answer card. Pairs move any wrong cards. Look hard at the daddy longlegs and the pill bug.
4. Event Judge 8 min. Put the 4 Station Sprint cards (and any specimens) around the table. Set the timer for 2 minutes per station and call 'Rotate!' each time. Pairs write answers in the Station Sprint table and cannot go back.
5. Both 3 min. Set the timer for 3 minutes. Answer the 2 'Think' questions. Put this page in your Buzzworthy divider.

Check your work: Ask your Event Captain for the answer card and check your Station Sprint and Think answers. Fix any you missed in a different color.

The 5 arthropod classes

Class	Body parts	Legs	Antennae	Examples
Arachnida	2	8	none	spiders, scorpions, ticks, mites, daddy longlegs
Chilopoda	many flat segments	1 pair on each segment	1 pair, long	centipedes (fast hunters with venom claws)
Diplopoda	many round segments	2 pairs on most segments	1 pair, short	millipedes (slow, eat dead plants, curl up)
Crustacea	usually 2	10 or more	2 pairs	crabs, shrimp, crayfish, barnacles, pill bugs
Insecta	3 (head, thorax, abdomen)	6	1 pair	beetles, ants, butterflies, grasshoppers

Station Sprint

Station	Class	Part 2 answer
1. Where does it like to live?		
2. How many legs? Antennae?		
3. Legs on each segment?		
4. What is attached to the thorax?		

Think

1. What 3 things do you count or look for to decide if an animal is an insect?

_____

_____

2. A pill bug looks like a bug. Why is it NOT an insect?

At home before next time: Arthropod hunt (15 minutes): look around your home or yard (under a rock, on a window, on a plant) or look at photos with an adult. Find 3 arthropods from at least 2 different classes. Log each one on a Field Observation page in your Bug Detective Field Journal (pages 8-15): sketch it, count legs and body parts, and write its class. That earns the Class Explorer badge. Do not pick up spiders or stinging insects. Bonus: do Pollinator Patrol (journal p. 17) on a sunny day while flowers are still blooming. Optional: watch the 'What is an Insect?' video from the NCSO page.

Insect Body Parts, Life Cycles, and 6 Big Orders

TODAY I CAN

- I can tell an insect from other arthropods by its 3 body parts, 6 legs, and 1 pair of antennae.
- I can label an insect's head, thorax, abdomen, antennae, compound eyes, mouthparts, legs, wings, and spiracles.
- I can explain complete metamorphosis (egg, larva, pupa, adult) and incomplete metamorphosis (egg, nymph, adult), and say which one an order uses.
- I can identify Coleoptera, Lepidoptera, Hymenoptera, Diptera, Neuroptera, and Siphonaptera from a photo or specimen.
- I can match mouthparts (chewing, piercing-sucking, siphoning, sponging) to an insect and what it eats.

Most insects go through a big change as they grow, called metamorphosis (met-a-MOR-fo-sis). This page goes in your Ref 1 ID Binder.

Today's steps: run it yourselves (22 minutes)

You need: Mealworm cup (larvae, pupae, beetles), 12 order photo cards per pair, 6 order-name cards per pair, Lesson 2 answer card. Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Draw a simple insect in the 'Label the insect' box. Label every part from the word bank. Spiracles are tiny breathing holes along the sides.
2. Both 4 min. Set the timer for 4 minutes. Use hand lenses to find a larva, a pupa, and an adult beetle in the mealworm cup. Keep the lid on. Read 'Two kinds of metamorphosis': which kind is this?
3. Partner A 2 min. Set the timer for 2 minutes. Read the 'Mouthparts' box. Act out one mouthpart (chewing, sipping, sponging, or poking) and Partner B names it and an insect that has it. Switch.
4. Both 7 min. Set the timer for 7 minutes. Lay out the 6 order-name cards. Match the 12 photos to them using 'Order clues'. Hint: ptera means wing, so each name tells you about the wings.
5. Both 3 min. Set the timer for 3 minutes. Answer the 4 'Check yourself' questions.
6. Event Captain 2 min. Read the order match and question answers from the answer card. Pairs fix mistakes, then file the page in Ref 1.

Check your work: Ask your Event Captain for the answer card and check your answers. Fix any you missed in a different color.

Label the insect

Draw a simple insect here: 3 body parts, 6 legs on the middle part, 2 antennae, and wings. Label every part from the word bank.

Word bank

head **thorax** **abdomen** **antenna** **compound eye** **mouthparts** **leg** **wing** **spiracle**

Two kinds of metamorphosis

Complete	egg, larva (a worm-like baby, like a caterpillar or grub), pupa (a resting stage), adult. The young look nothing like the adult.

Incomplete	egg, nymph, adult. A nymph looks like a small adult without full wings. It molts (sheds its skin) as it grows.
None	Young look like tiny adults and just molt, even as adults (silverfish, bristletails).

Mouthparts

Chewing	jaws that bite and grind: beetles, grasshoppers
Siphoning	a coiled tube (proboscis) to sip nectar: butterflies, moths
Sponging	a spongy pad that mops up liquid: house flies
Piercing-sucking	a needle-like beak that pokes and sucks: mosquitoes, true bugs

Order clues: complete metamorphosis (ptera = wing)

Coleoptera (sheath wing)	Beetles, ladybugs, fireflies, weevils. Hard front wings (elytra) meet in a straight line down the back. Chewing mouth. The biggest order of all.
Lepidoptera (scale wing)	Butterflies and moths. Wings covered in tiny scales. Coiled proboscis. Larva is a caterpillar.
Hymenoptera (membrane wing)	Bees, wasps, ants, sawflies. Thin waist. 2 pairs of clear wings; the back pair is smaller. Many females can sting.
Diptera (two wing)	Flies, mosquitoes, gnats. Only 1 pair of wings; tiny knobs called halteres help them balance. Larva is a maggot.
Neuroptera (nerve wing)	Lacewings and antlions. Lacy wings full of veins, held like a roof. Larvae hunt aphids and other insects.
Siphonaptera (tube, no wing)	Fleas. No wings. Body flat side to side. Huge jumping legs. Adults drink blood from mammals and birds.

Check yourself

1. List the 4 stages of complete metamorphosis in order.

2. How can you tell a fly from a bee?

3. Which order has a coiled tube mouth? What kind of mouthpart is it?

4. A tiny insect with no wings jumps off a dog. Which order? Give 2 clues.

At home before next time: Do Independent Lab I1 (Insect Relationships) before Lesson 3 on Dec 11. Also make 6 order cards for your ID Binder: on 6 index cards write one order name each (Coleoptera, Lepidoptera, Hymenoptera, Diptera, Neuroptera, Siphonaptera), draw or glue a picture, and write 2 clues. Field Journal: find an insect that does complete metamorphosis (any life stage), log it on an observation page, and fill in today's Mealworm Log count (p. 16). That earns the Metamorphosis Master badge. Museum trip tomorrow (Sat Nov 7, meet 10 am at the Daily Planet Cafe; 10:45 tour of the insect and arthropod research lab): bring your journal and fill in the Museum Day page (p. 20).

Hoppers, Bugs, Roaches, and Hiders

TODAY I CAN

- I can explain complete metamorphosis (egg, larva, pupa, adult) and incomplete metamorphosis (egg, nymph, adult), and say which one an order uses.
- I can identify Orthoptera, Hemiptera, Blattodea, Dermaptera, Phasmatodea, and Psocodea from a photo or specimen.
- I can match mouthparts (chewing, piercing-sucking, siphoning, sponging) to an insect and what it eats.
- I can describe insect habitats (including ponds and streams) and adaptations like camouflage, mimicry, and warning colors.

These 6 orders all have incomplete metamorphosis: egg, nymph, adult. This page goes in your Ref 1 ID Binder.

Today's steps: run it yourselves (27 minutes)

You need: 12 order photo cards per pair, Camouflage game bag (paper floor + 30 strips), 3 station practice cards, Lesson 3 answer card. Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Take turns reading 'Order clues: incomplete metamorphosis' out loud, one order each. Point to the best clue in each row.
2. Both 7 min. Set the timer for 7 minutes. Sort the 12 photo cards into the 6 orders. Write the order and the clue you used in the 'Order detective' table.
3. Team 6 min. Read 'Adaptations for staying alive'. The Event Judge spreads all 30 strips on the paper floor while everyone looks away. Each student is a bird: 15 seconds to pick up strips one at a time. Count which colors got caught and talk about why.
4. Event Judge 6 min. Put the 3 station cards around the table. Set the timer for 2 minutes per station and call 'Rotate!'. Pairs write both parts in the 'Station practice' table.
5. Both 4 min. Set the timer for 4 minutes. Answer the 2 'Think' questions, check with the answer card, and file the page in Ref 1.

Check your work: Ask your Event Captain for the answer card and check your answers. Fix any you missed in a different color.

Order clues: incomplete metamorphosis

Orthoptera (straight wing)	Grasshoppers, crickets, katydids. Big back legs for jumping. Chewing mouth. Males 'sing' by rubbing body parts together.
Hemiptera (half wing)	True bugs: stink bugs, cicadas, aphids, water striders. Piercing-sucking beak. Many have front wings half leathery, half clear, making an X on the back.
Blattodea	Cockroaches and termites. Roaches: flat, fast, shield over the head, long antennae. Termites: pale, soft, straight antennae, thick waist; eat wood with help from gut microbes.
Dermaptera (skin wing)	Earwigs. Pincers (cerci) at the end of the abdomen. Short leathery front wings. Like damp, dark places.
Phasmatodea	Stick insects and leaf insects. Look like twigs or leaves. Eat leaves. Move slowly and sway.
Psocodea	Lice and booklice. Lice: tiny, flat, no wings, claws that grip hair; they are parasites. Booklice eat mold on old paper.

Adaptations for staying alive

Adaptation	a body part or behavior that helps a living thing survive
Camouflage	blending in with the background (stick insect, leaf insect)
Warning colors	bright colors that say 'I taste bad or I sting' (milkweed bug, wasp)
Mimicry	looking like a different, dangerous animal (a hoverfly looks like a bee)

Order detective

Photo	Order	Clue I used
grasshopper or cricket		
stink bug or cicada		
cockroach or termite		
earwig		
stick or leaf insect		
louse or booklouse		

Station practice

Station	Order	Part 2
1		
2		
3		

Think

1. Explain camouflage, warning colors, and mimicry in your own words.

2. Which order includes termites? How do they digest wood?

At home before next time: Do Independent Lab I2 (Insects, Health, and Money) before Lesson 4 on Jan 29. Also make 6 more order cards (Orthoptera, Hemiptera, Blattodea, Dermaptera, Phasmatodea, Psocodea) with a picture and 2 clues each. Shuffle all 12 cards and time yourself naming them. Write your best time here: ____ seconds. Field Journal: with an adult, do a Log Roller Survey (p. 18) and look for a hider using camouflage. That earns the Log Roller badge. Bring your cards and journal next time.

Water Insects, Ancient Insects, and the Key

TODAY I CAN

- I can name the 5 arthropod classes (Arachnida, Chilopoda, Crustacea, Diplopoda, Insecta) and give an example of each.
- I can explain complete metamorphosis (egg, larva, pupa, adult) and incomplete metamorphosis (egg, nymph, adult), and say which one an order uses.
- I can identify Odonata, Ephemeroptera, Trichoptera, Archaeognatha, and Zygentoma from a photo or specimen.
- I can use a dichotomous key to find an arthropod's class and insect order.
- I can describe insect habitats (including ponds and streams) and adaptations like camouflage, mimicry, and warning colors.

A dichotomous (die-KOT-o-mus) key gives you 2 choices at each step (step 14 has 3). Pick the one that fits, then go to the next number. This key goes in your Ref 1 ID Binder.

Today's steps: run it yourselves (27 minutes)

You need: 10 water and ancient insect photo cards per pair, 5 habitat cards, 8 key practice cards per pair (numbered 1-8), Caddisfly case tray and masking tape, Lesson 4 answer card. Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Read the Odonata, Ephemeroptera, and Trichoptera rows of 'Order clues'. Find each adult and its young in the photo cards. Answer 'Think' questions 1 and 2.
2. Both 5 min. Set the timer for 5 minutes. Caddisfly case: wrap masking tape around a pencil, sticky side OUT. Roll it in the gravel, lentils, and twig bits, then slide it off. Talk: how does a case help a soft larva?
3. Both 3 min. Set the timer for 3 minutes. Read the Archaeognatha and Zygentoma rows. Compare the bristletail and silverfish photos, then answer 'Think' question 3.
4. Both 9 min. Set the timer for 9 minutes. Key out card 1 together: start at step 1, read a AND b, pick one, go to the next number. Then key out cards 2-8. Check 'Tricky ones'. Write answers in the 'Key practice' table.
5. Both 3 min. Set the timer for 3 minutes. Lay out the 5 habitat cards. Put every photo card on the habitat where it lives.
6. Event Captain 3 min. Read the key practice and Think answers from the answer card. Pairs fix mistakes and file the page in Ref 1.

Check your work: Ask your Event Captain for the answer card and check your answers. Fix any you missed in a different color.

Order clues: water insects and ancient insects

Odonata	Dragonflies and damselflies. Huge eyes, very long thin abdomen, 4 strong wings. Dragonflies rest with wings open flat; damselflies rest with wings closed over the back. Nymphs live in water. Incomplete metamorphosis.
Ephemeroptera	Mayflies. Wings held straight up over the back, 2 or 3 long tail threads. Adults live only a day or two and do not eat. Nymphs live in clean water. Incomplete.
Trichoptera (hair wing)	Caddisflies. Hairy wings held like a tent, long antennae; look like moths. Larvae live in water and build cases of pebbles or sticks. Complete.
Archaeognatha	Jumping bristletails. No wings, humped back, big eyes that touch, 3 tail threads (middle one longest). Can jump. No metamorphosis.
Zygentoma	Silverfish and firebrats. No wings, flat, silvery scales, 3 tail threads, small eyes far apart. Live in homes; eat paper glue and starch. No metamorphosis.

Arthropod key (use for adults)

1. 1a. 8 legs, no antennae: ARACHNIDA. 1b. Not 8 legs: go to 2.
2. 2a. 6 legs and 3 body parts: INSECTA, go to 5. 2b. 10 or more legs: go to 3.
3. 3a. 2 pairs of antennae, hard shell, most live in water: CRUSTACEA. 3b. 1 pair of antennae, long body of many segments: go to 4.
4. 4a. 1 pair of legs per segment, flat, fast: CHILOPODA. 4b. 2 pairs of legs per segment, round, slow: DIPLOPODA.
5. 5a. Has wings: go to 9. 5b. No wings: go to 6.
6. 6a. 3 long tail threads: go to 7. 6b. No tail threads: go to 8.
7. 7a. Humped back, big eyes that touch: ARCHAEOGNATHA. 7b. Flat and silvery, small eyes far apart: ZYGENTOMA.
8. 8a. Flat side to side, big jumping legs: SIPHONAPTERA. 8b. Flat top to bottom, tiny, claws grip hair: PSOCODEA.
9. 9a. Only 1 pair of wings: DIPTERA. 9b. 2 pairs of wings: go to 10.
10. 10a. Front wings hard or leathery: go to 11. 10b. Front wings thin (clear, scaly, or hairy): go to 15.

11. 11a. Pincers at end of abdomen: DERMAPTERA. 11b. No pincers: go to 12.
12. 12a. Hard front wings meet in a straight line down the back: COLEOPTERA. 12b. Not a straight line: go to 13.
13. 13a. Sucking beak, wings make an X on the back: HEMIPTERA. 13b. Chewing mouth: go to 14.
14. 14a. Big jumping back legs: ORTHOPTERA. 14b. Looks like a stick or leaf: PHASMATODEA. 14c. Flat and oval, shield over head: BLATTODEA.
15. 15a. Wings covered in scales, coiled mouth tube: LEPIDOPTERA. 15b. No scales: go to 16.
16. 16a. Hairy wings held like a tent: TRICHOPTERA. 16b. Clear wings: go to 17.
17. 17a. Very long thin abdomen, huge eyes, tiny antennae: ODONATA. 17b. Not: go to 18.
18. 18a. Wings held straight up, 2 or 3 long tail threads: EPHEMEROPTERA. 18b. No tail threads: go to 19.
19. 19a. Thin waist, back wings smaller than front: HYMENOPTERA. 19b. Lacy net of veins, wings held like a roof: NEUROPTERA.

Tricky ones

No wings but not in steps 6-8	worker ants (Hymenoptera, thin waist), termite workers (Blattodea), many stick insects (Phasmatodea)
Clear wings but a beak	cicadas and aphids are Hemiptera
Looks like a bee	a hoverfly has only 1 pair of wings: Diptera
Looks like an insect	daddy longlegs (8 legs): Arachnida; pill bug: Crustacea

Key practice

Card	Class	Order (if insect)
1		
2		
3		
4		
5		
6		
7		
8		

Think

1. Which 3 orders have young that live underwater?

2. You find lots of mayfly and caddisfly young in a creek. What does that tell you?

3. How can you tell a silverfish from a jumping bristletail?

At home before next time: Make your last 5 order cards (Odonata, Ephemeroptera, Trichoptera, Archaeognatha, Zygentoma). Have someone pick 5 of your 17 cards; use the key to find each order without looking. Write a habitat word on each card (water, soil, plants, homes, or animals) and put all 17 in sheet protectors in Ref 1. Field Journal: use the pocket key (p. 7) on 3 bugs and write the steps on observation pages (Key Master badge). Bonus: Water Watch (p. 19) at a creek with an adult, or with photos.

Practice Tournament

TODAY I CAN

- I can name the 5 arthropod classes (Arachnida, Chilopoda, Crustacea, Diplopoda, Insecta) and give an example of each.
- I can identify Coleoptera, Lepidoptera, Hymenoptera, Diptera, Neuroptera, and Siphonaptera from a photo or specimen.
- I can identify Orthoptera, Hemiptera, Blattodea, Dermaptera, Phasmatodea, and Psocodea from a photo or specimen.
- I can identify Odonata, Ephemeroptera, Trichoptera, Archaeognatha, and Zygentoma from a photo or specimen.
- I can use a dichotomous key to find an arthropod's class and insect order.
- I can explain pollination, decomposition, food webs, competition, and symbiosis (mutualism, commensalism, parasitism) with insect examples.
- I can name arthropods that spread disease (mosquitoes, fleas, lice, ticks) and ways people stay safe.
- I can find an answer in our 2 reference binders quickly and finish each station before time is called.

Practice tournament! Write the name first (class and/or order), then answer Part 2. When time is called, move on.

Today's steps: run it yourselves (27 minutes)

You need: 8 station cards (photo + question, 1-8), Lesson 5 answer card. Set the timer for each step.

1. Event Judge 1 min. Set out the 8 stations in number order. Read aloud: '3 minutes per station. When time is called, move to the next number. You cannot go back. Use your binders.' Pairs start at Stations 1 and 5.
2. Event Judge 12 min. Set the timer for 3 minutes each round for 4 rounds. Call 'One minute left' and 'Rotate!'. Pairs write the name in Part 1 and answer Part 2 on the 'Answer sheet'.
3. Event Judge 12 min. Keep going for 4 more rounds so every pair visits all 8 stations. No hints: use your binders and the key.
4. Event Captain 2 min. Read the answers from the answer card. Pairs score their sheet and fill in 'My review plan'.

Check your work: Ask your Event Captain for the answer card and score your answer sheet. Circle the 2 stations you missed and write the binder page to review.

Answer sheet

Station	Part 1: class and/or order	Part 2 answer
1		
2		
3		
4		
5		
6		
7		
8		

Before the tournament

- [ ] I can name all 5 classes and all 17 orders from a picture.
- [ ] My 2 binders have tabs and every page is attached to the rings.
- [ ] I know where the key, the Tricky Ones box, and the health page are.
- [ ] I bring pencils and 2 hand lenses.

My review plan

1. Which 2 stations did I miss? What page will I study?

_____

_____

At home before next time: Review your 17 order cards and the 5 classes every other day until the tournament on Sat Mar 13 (10 minutes each time). Work on the stations you missed today. Add any missing facts to your binders in pencil. Field Journal: do the Mystery Bug Challenge (p. 21) in under 2 minutes with 6 or more right (Speed Spotter badge), and spot or study 10 of the 17 orders on the Order Collector pages (p. 5-6) before Mar 13 (Order Collector badge).

Lab I1: Who Needs Whom? Insect Relationships

THIS LAB HELPS ME

- I can explain pollination, decomposition, food webs, competition, and symbiosis (mutualism, commensalism, parasitism) with insect examples.
- I can explain how climate change affects insects and ticks (warmer winters, moving ranges, bad timing with flowers).

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Insects are linked to every living thing around them. This page starts your Ref 2 Ecology and Economics Binder, behind the Relationships tab.

Steps (29 minutes)

You need: This handout, Pencil and colored pencils. Set the timer for each step.

1. Me 4 min. Set a timer for 4 minutes. Read the 'Relationships' box out loud. For each word, think of one example of your own and say it.
2. Me 6 min. Set a timer for 6 minutes. Read each story in 'Name the relationship' and circle the best word. Look back at the box when you are stuck.
3. Me 7 min. Set a timer for 7 minutes. In 'Build a food web', write the 7 names in the box. Draw an arrow from each food to the animal that eats it. Arrows point to the eater.
4. Me 4 min. Set a timer for 4 minutes. Read 'Climate change and insects' out loud, to yourself or to someone at home.
5. Me 5 min. Set a timer for 5 minutes. Answer the 3 'Think' questions in full sentences.
6. Me 3 min. Check with the Lab I1 answer card in the Open Lab bin (or ask your Event Captain at Lesson 3). Fix mistakes in a different color.

Check your work: Use the Lab I1 answer card from the Open Lab bin, or bring the page to Lesson 3 and check it with your Event Captain.

Relationships

Pollination	carrying pollen from flower to flower so plants make seeds and fruit (bees, butterflies, moths, flies, beetles)
Decomposer	breaks down dead plants, animals, and waste into soil (dung beetles, termites, millipedes, pill bugs, fly maggots)
Food web	who eats whom: plants, then plant eaters (aphids), then predators (lacewing larvae, dragonflies, spiders), then birds, frogs, bats, fish
Competition	two living things need the same food or space (honey bees and native bees visiting the same flowers)
Mutualism	both win: bees get nectar, flowers get pollinated; ants guard aphids and eat their sweet honeydew; termites and their gut microbes
Commensalism	one wins, the other is not helped or hurt: tiny mites ride on beetles to new places
Parasitism	one wins, the host loses: fleas, lice, ticks, and mosquitoes drink blood

Name the relationship

1. Ants guard aphids and eat the sweet honeydew the aphids make. The aphids stay safe.

○ pollination ○ decomposer ○ competition ○ mutualism ○ commensalism ○ parasitism

2. A tick drinks a deer's blood.

○ pollination ○ decomposer ○ competition ○ mutualism ○ commensalism ○ parasitism

3. Tiny mites ride on a beetle to a new log. The beetle is not helped or hurt.

○ pollination ○ decomposer ○ competition ○ mutualism ○ commensalism ○ parasitism

4. Honey bees and bumble bees visit the same flowers for nectar.

○ pollination ○ decomposer ○ competition ○ mutualism ○ commensalism ○ parasitism

5. A dung beetle rolls away animal waste and buries it.

○ pollination ○ decomposer ○ competition ○ mutualism ○ commensalism ○ parasitism

6. A butterfly carries pollen from one flower to another.

○ pollination ○ decomposer ○ competition ○ mutualism ○ commensalism ○ parasitism

Build a food web

Write: milkweed plant, aphid, ladybug, spider, bluebird, dead leaves, millipede. Draw arrows from each food to its eater.

Climate change and insects

Warmer winters	more pests and ticks survive; mosquitoes and ticks spread north
Moving ranges	insects move north or up mountains to stay cool
Timing	flowers may bloom before pollinators come out, so both suffer

Think

1. If all the ladybugs in a garden disappeared, what would happen to the aphids and the plants?

2. How could warmer winters change where ticks and mosquitoes live?

3. Why is it bad for both if flowers bloom before the bees come out?

Bring to the next meeting: Bring this page to Lesson 3 and file it in Ref 2 behind the Relationships tab.

Lab I2: Insects, Health, and Money

THIS LAB HELPS ME

- I can name arthropods that spread disease (mosquitoes, fleas, lice, ticks) and ways people stay safe.
- I can explain how insects help people (food, honey, silk, dyes, medicine) and harm people (crop pests, house pests), and what integrated pest management means.
- I can find an answer in our 2 reference binders quickly and finish each station before time is called.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Some insects make food, cloth, and dye. Some spread disease or eat crops. This page goes in your Ref 2 Ecology and Economics Binder.

Steps (30 minutes)

You need: This handout and 1 blank sheet, Ref 2 binder, tab dividers, sheet protectors. Set the timer for each step.

1. Me 3 min. Set a timer for 3 minutes. Read 'Public health (disease)' out loud.
2. Me 3 min. Set a timer for 3 minutes. Read 'Economics: insects and money' out loud.
3. Me 7 min. Set a timer for 7 minutes. Fill in 'Helper or Harmer?'. Write helps, harms, or both, and the product or problem. Use the two boxes above.
4. Me 5 min. Set a timer for 5 minutes. Answer the 'Think' questions. For the pest plan, start with prevent.
5. Me 5 min. Set a timer for 5 minutes. On the blank sheet, make your own Ref 2 page: pick one insect (like the honey bee or the mosquito), draw it, and write 5 facts about how it helps or harms people.
6. Me 4 min. Set a timer for 4 minutes. Set up Ref 2 with the 'Ref 2 set up' checklist. Then time how fast you can find the answers to the 3 speed questions.
7. Me 3 min. Check with the Lab I2 answer card in the Open Lab bin (or ask your Event Captain at Lesson 4). Fix mistakes in a different color.

Check your work: Use the Lab I2 answer card from the Open Lab bin, or bring the page to Lesson 4 and check it with your Event Captain.

Public health (disease)

Mosquitoes (Diptera)	malaria, West Nile virus, Zika, dengue. Stay safe: dump standing water, use screens and repellent.
Fleas (Siphonaptera)	carried the plague. Treat pets for fleas.
Body lice (Psocodea)	can spread typhus. Head lice are itchy but do not spread disease.
Ticks (Arachnida)	Lyme disease, Rocky Mountain spotted fever. Check your skin after being outside.
Flies (Diptera)	house flies carry germs on their feet to food.

Economics: insects and money

Food	honey; crickets and mealworms eaten as protein; about 1 in 3 bites of our food depends on pollinators
Products	silk (silkworm moth), beeswax, shellac (lac insect), red dye carmine (cochineal)
Medicine	clean maggots are used to clean wounds; honey is used on some burns and cuts
Pests	crop pests (Japanese beetles, aphids, caterpillars), house pests (termites, cockroaches)
Integrated pest management	prevent pests first, check often, use the least harmful fix

Helper or Harmer?

Insect	Helps, harms, or both?	Product or problem
honey bee		

Insect	Helps, harms, or both?	Product or problem
silkworm moth		
cochineal		
lac insect		
dung beetle		
cricket		
mosquito		
flea		
body louse		
tick		
Japanese beetle		
termite		

Think

1. Which insect order spreads malaria?

2. A tick is on this list. Is it an insect? How do you know?

3. Mosquitoes are biting in your backyard. Write an integrated pest management plan with 3 steps, starting with prevent.

Ref 2 set up (every page on the rings)

- [ ] Tabs: Relationships, Health, Helpers and Pests, Climate
- [ ] Lab I1 page behind Relationships
- [ ] This page behind Health
- [ ] My own insect page behind Helpers and Pests
- [ ] Speed find: Which disease do ticks spread? Which insect makes a red dye? What did fleas spread? My time: ___ seconds

Bring to the next meeting: Bring your Ref 2 binder to Lesson 4 with this page and your own insect page inside.